

MVLU COLLEGE

AI PRACTICAL NO. 5

AIM:Decision Tree Learning

- Implement the Decision Tree Learning algorithm to build a decision tree for a given dataset.
- Evaluate the accuracy and effectiveness of the decision tree on test data.
- Visualize and interpret the generated decision tree.

#CODE:

```
# PRACTICAL 5: DECISION TREE LEARNING

# Import libraries

import pandas as pd

import matplotlib.pyplot as plt

from google.colab import files

from sklearn.model_selection import train_test_split

from sklearn.compose import ColumnTransformer

from sklearn.pipeline import Pipeline

from sklearn.preprocessing import OneHotEncoder

from sklearn.impute import SimpleImputer

from sklearn.tree import DecisionTreeClassifier, plot_tree

from sklearn.metrics import accuracy_score, classification_report,
confusion_matrix

# Upload dataset

uploaded = files.upload()

# Read dataset

df = pd.read_csv("Weather.csv")

# Display first 5 rows

print("First 5 rows of dataset:")
```

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```
print(df.head())

print("\nDataset Shape:")

print(df.shape)

print("\nColumns:")

print(df.columns)

X = df.drop(columns=["RainTomorrow", "RISK_MM"])

y = df["RainTomorrow"]

# Identify numerical and categorical columns

numeric_columns = X.select_dtypes(exclude="object").columns

categorical_columns = X.select_dtypes(include="object").columns

# Preprocessing

numeric_transformer = Pipeline(

    steps=[

        ("imputer", SimpleImputer(strategy="median"))

    ]

)

categorical_transformer = Pipeline(

    steps=[

        ("imputer", SimpleImputer(strategy="most_frequent")),

        ("encoder", OneHotEncoder(handle_unknown="ignore"))

    ]

)

preprocessor = ColumnTransformer(
```

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```
transformers=[

    ("num", numeric_transformer, numeric_columns),

    ("cat", categorical_transformer, categorical_columns)

]

)

# Decision Tree model

decision_tree = DecisionTreeClassifier(

    criterion="entropy",

    max_depth=4,

    random_state=42

)

model = Pipeline(

    steps=[

        ("preprocessor", preprocessor),

        ("classifier", decision_tree)

    ]

)

# Split dataset

X_train, X_test, y_train, y_test = train_test_split(

    X,

    y,

    test_size=0.25,

    random_state=42,
```

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```
        stratify=y
    )

    # Train model

    model.fit(X_train, y_train)

    # Prediction

    y_pred = model.predict(X_test)

    # Accuracy

    accuracy = accuracy_score(y_test, y_pred)

    print("\nDecision Tree Accuracy:")

    print(accuracy)

    print("\nAccuracy Percentage:")

    print(round(accuracy * 100, 2), "%")

    print("\nConfusion Matrix:")

    print(confusion_matrix(y_test, y_pred))

    print("\nClassification Report:")

    print(classification_report(y_test, y_pred))

    # Get feature names after encoding

    feature_names = model.named_steps["preprocessor"].get_feature_names_out()

    # Get trained decision tree

    trained_tree = model.named_steps["classifier"]

    # Visualize Decision Tree

    plt.figure(figsize=(25, 12))

    plot_tree(
```

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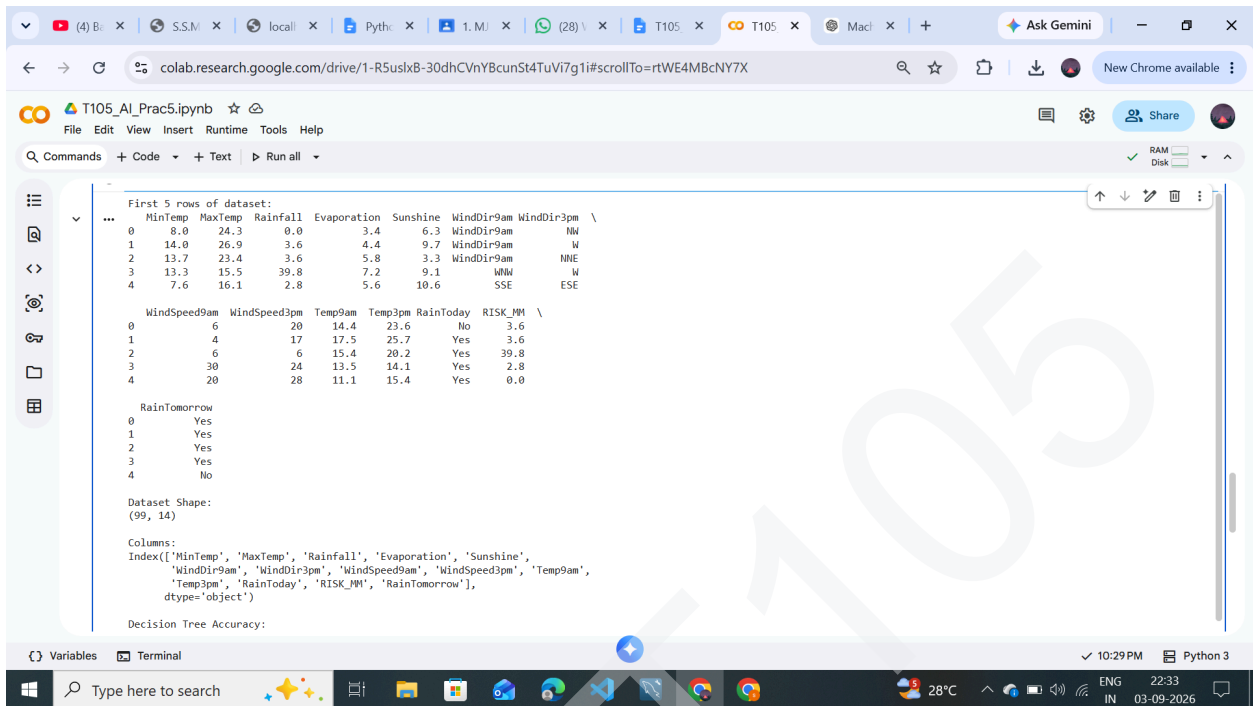
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```
trained_tree,  
  
feature_names=feature_names,  
  
class_names=[str(x) for x in trained_tree.classes_],  
  
filled=True,  
  
rounded=True,  
  
fontsize=8  
  
)  
  
plt.title("Decision Tree for Rain Tomorrow Prediction")  
  
plt.show()  
  
# Sample predictions  
  
results = pd.DataFrame({  
  
    "Actual": y_test.values,  
  
    "Predicted": y_pred  
  
})  
  
print("\nActual vs Predicted:")  
  
print(results.head(10))
```

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#OUTPUT:



First 5 rows of dataset:

	MinTemp	MaxTemp	Rainfall	Evaporation	Sunshine	WindDir9am	WindDir3pm
0	8.0	24.3	0.0	3.4	6.3	WindDir9am	NW
1	14.0	26.9	3.6	4.4	9.7	WindDir9am	W
2	13.7	23.4	3.6	5.8	3.3	WindDir9am	NNE
3	13.3	15.5	39.8	7.2	9.1	WNW	W
4	7.6	16.1	2.8	5.6	10.6	SSE	ESE

WindSpeed9am WindSpeed3pm Temp9am Temp3pm RainToday RISK_MM \

	WindSpeed9am	WindSpeed3pm	Temp9am	Temp3pm	RainToday	RISK_MM
0	6	20	14.4	23.6	No	3.6
1	4	17	17.5	25.7	Yes	3.6
2	6	6	15.4	20.2	Yes	39.8
3	30	24	13.5	14.1	Yes	2.8
4	20	28	11.1	15.4	Yes	0.0

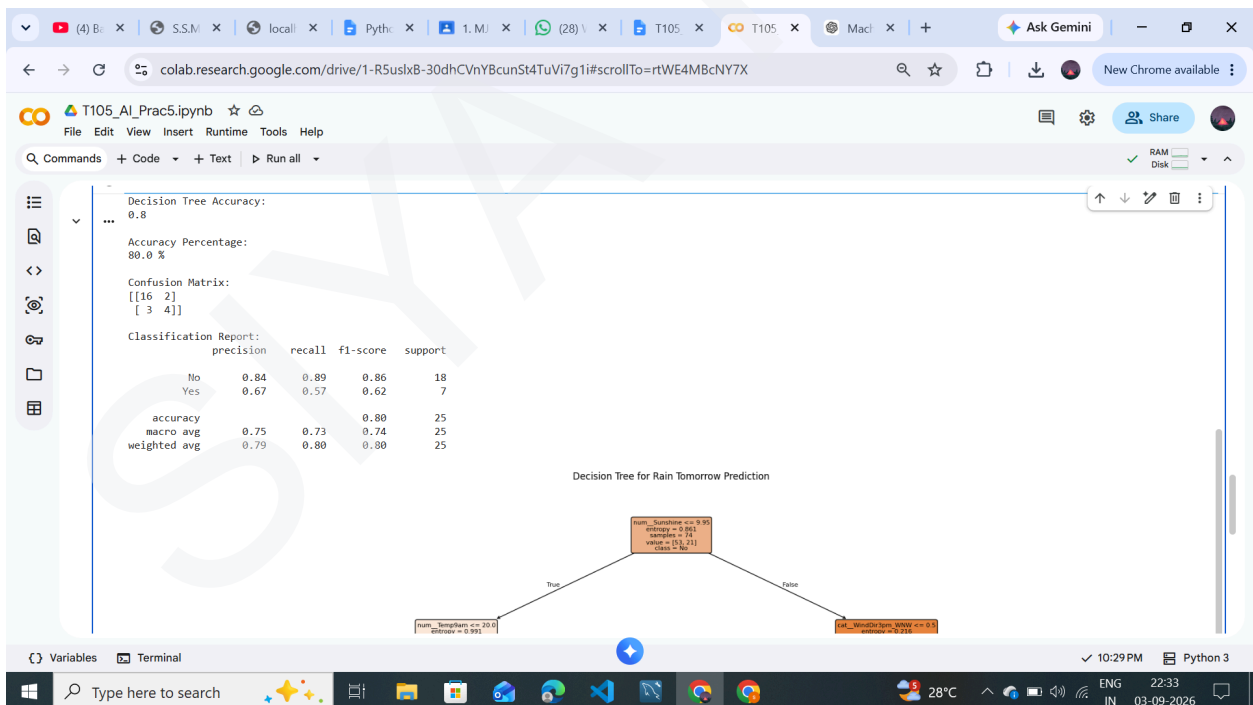
RainTomorrow

	RainTomorrow
0	Yes
1	Yes
2	Yes
3	Yes
4	No

Dataset Shape:
(99, 14)

Columns:
Index(['MinTemp', 'MaxTemp', 'Rainfall', 'Evaporation', 'Sunshine', 'WindDir9am', 'WindDir3pm', 'WindSpeed9am', 'WindSpeed3pm', 'Temp9am', 'Temp3pm', 'RainToday', 'RISK_MM', 'RainTomorrow'], dtype='object')

Decision Tree Accuracy:



Decision Tree Accuracy:
0.8

Accuracy Percentage:
80.0 %

Confusion Matrix:
[[16 2]
 [3 4]]

Classification Report:

	precision	recall	f1-score	support
No	0.84	0.89	0.86	18
Yes	0.67	0.57	0.62	7
accuracy			0.80	25
macro avg	0.75	0.73	0.74	25
weighted avg	0.79	0.80	0.80	25

Decision Tree for Rain Tomorrow Prediction

```
graph TD
    Root["Rain_Sunshine <= 9.50  
entropy = 0.981  
samples = 25  
value = [16, 9]  
class = No"]
    Root -- True --> Left["Rain_Temp9am <= 20.0  
entropy = 0.915"]
    Root -- False --> Right["Rain_WindDir9am_Split <= 0.0"]
```

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Decision Tree for Rain Tomorrow Prediction

